Lesson plan · for the teacher

Deciding with data

Data science · Level 4 · Lesson 4

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Write a clear data question.
- Design a fair trial.
- Decide with numbers and voices.

Activity: Plan a fair trial

1. Warm-up (5 min): Ask colleagues for a school rule nobody has tested. How do we know it works?
2. Teach (8 min): Walk through the question, measures, trial and comparison, and work out the effect together.
3. Plan (15 min): Groups choose a real question and fill a one-page trial plan.
4. Share (7 min): Each group pitches its plan in one minute; others ask whose voices are missing.
5. Wrap-up (5 min): Ask two colleagues to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- What is your comparison group?
- Whose views will you ask?
- How will you share a result that disappoints?

Exit ticket

- Trial: 20 → 14. Comparison: 19 → 17. Work out the effect.

Try it at school tomorrow
Bring one school question to a staff meeting, agree on what to measure before collecting anything, and plan a small trial with a fair comparison group.

Practice questions · for you

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1 **The data shows a later start cuts lateness. Who should make the final decision?**

◯ The data ◯ The school's people, using data and voices ◯ The chart

2 **Put the steps of a data-led decision in order.**

☐ Run a fair trial

☐ Decide and share the results

☐ Ask the people affected how the trial went

☐ Write one clear question

☐ Agree on what to measure

3 **Work out the effect of the later start: the trial classes' drop in lateness minus the other classes' drop. Print it.**

```
trial_before = 12
trial_after = 5
other_before = 11
other_after = 10
effect = trial_before - trial_after
print(effect)
```

Write your program:

4 **Why decide what to measure before collecting data?**

◯ Because charts need it ◯ To save paper ◯ So nobody picks the numbers they like afterwards

5 **If the numbers clearly favour a change, parents' and pupils' views no longer matter.**

◯ True ◯ False

✎ **Which decision at your school was made on a hunch that a small fair trial could test?**

Check your answers

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1 **The school's people, using data and voices**

Data informs; people weigh it and decide.

2 **Top to bottom: 3, 5, 4, 1, 2**

Fix the question and measures first, so the evidence can't be bent to fit a wish.

3

```
trial_before = 12
trial_after = 5
other_before = 11
other_after = 10
effect = (trial_before - trial_after) - (other_before - other_after)
print(effect)
```

Lateness fell by 7 in the trial classes and by 1 anyway in the others, so the change itself explains about 6 fewer late arrivals a week.

4 **So nobody picks the numbers they like afterwards**

Fixing measures first keeps the trial honest.

5 **False**

Voices reveal costs the numbers miss; people weigh both.